Gursimran Singh

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Research Interests

In-network computation and data-plane programming (P4); programmable data planes for network security; network and Internet measurement; network security and censorship measurement.

Education

Oklahoma State University — Stillwater, OK
Ph.D. in Computer Science; Advisor: Dr. H. B. Acharya — *Expected 2028*

Rochester Institute of Technology — Rochester, NY
Ph.D. coursework, Computer and Information Science; GPA 3.80/4.0 — *Transferred*

State University of New York at Buffalo — Buffalo, NY
M.S., Computer and Information Science; GPA 3.52/4.0 — *2021*

Publications

Programmable Data Planes for Network Security — NeTCoM, July 2025
- Position paper systematizing recent advances in security applications built on programmable switches.

POSTER: A Cyberspace Study of the Russia–Ukraine War — ACM ASIACCS, July 2023
- Measurement findings on the conflict's impact on Internet resilience.

Research Experience

In-Network Computation Using Data-Plane Programming in P4 — 2021 – Present
Oklahoma State University / RIT
- Developing high-performance packet processors comparable to application-layer firewalls and network intrusion detection systems.
- Implementing advanced algorithms for packet analysis and filtering in P4 on the Intel Tofino (TNA) platform.

Internet Censorship Measurement — 2021 – Present
Oklahoma State University / RIT
- Designed and implemented tools for detecting and measuring Internet censorship across multiple countries.
- Published measurement findings on the Russia–Ukraine conflict's impact on Internet resilience (ASIACCS 2023).

Teaching Experience

Adjunct Instructor, Rochester Institute of Technology — 2022, 2024 – 2025
CSEC-140 Intro to Cybersecurity; CSCI-455 Principles of Cybersecurity; GCIS-123 Software Dev I
- Delivered lectures and hands-on virtual labs covering authentication, data security, risk management, network and system security, application security, and human factors.
- Taught computational problem solving in Python: searching and sorting, file I/O, lists, classes, and dictionaries.

Teaching Assistant, Rochester Institute of Technology — 2023 – 2024
CSCI-243 Mechanics of Programming; CSEC-759 Networking & Distributed Systems Security

Teaching Assistant, Oklahoma State University — Spring 2026
Distributed and Cloud Computing

Academic & Professional Service

Technical Program Committee member, ACM S3 2026 Workshop (invited and accepted) — 2026
Shadow Technical Program Committee member, ACM CoNEXT 2026 — 2026
President, CS Graduate Student Association (CSGSA), Oklahoma State University — Current
Council Member, Doctoral Student Association (DSA), RIT — 2025
Vice-President, CSEGSA, University at Buffalo — 2020 – 2021
Mentorship: 3 master's and 2 undergraduate students, research and capstone projects — 2022 – 2025

Selected Projects

Reliable transport protocols and TCP chat system | *C/C++*
- Implemented Alternating-Bit, Go-Back-N, and Selective-Repeat protocols and a multi-client TCP chat system.

Task offloading for QoS optimization with a dynamic ANN | *C, C++, CUDA*
- Prototype that adapts at run time to query parameters (max-hops, query volume) to optimize QoS.

Core OS components in Pintos | *C*
- Kernel-level threading and system-call interfaces.

Technical Skills

Languages: C, C++, P4, Python, Java, Shell scripting

Networking & Security: Tofino Native Architecture (TNA), Scapy, Wireshark, Zmap, Nmap, Aircrack-ng, Kali Linux, OllyDbg, IDA

ML & Data: Scikit-learn, Keras, NumPy, Pandas

Certifications

KLCP and OSWP, Offensive Security 2023
Intel Tofino / P4: ICA-XFG-101 ($P4_{16}$, Tofino, SDE); ICA-XFG-201–206 (Advanced TNA) 2023 – 2024
Google Cloud: Intro to Generative AI, Responsible AI, and LLMs 2023